

# Companion to Bird & de Moor, Algebra of Programming

## §6. Recursive Programs

### §6.1. Digits of a number

(6.1a)

**$\text{val}^\circ = \text{embed}^\circ \text{wrap} \cup \text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{snoc}$**

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

$(\text{wrap}^\circ \text{embed})^\circ \cup (\text{snoc}^\circ \neg \text{Digit}(\text{val}) \text{op})^\circ \text{embed}^\circ \text{wrap} \cup \text{op}^\circ \neg \text{Digit}(\text{val})^\circ \text{snoc} \text{embed}^\circ \text{wrap} \cup \text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{snoc}$

(a) (b) (c) (d) (e)

(a) the converse of **val**, which reads digits as a number

(b) **val** undoes **wrap** or **snoc**, reads the front with **val** and combines by **embed** or **op**, so its converse runs that backwards

(c) undoing **wrap** or **snoc** first gives two cases: a decimal and its last digit read by **val** then **op**, or one digit read by **embed**

(d) the converse of each case is its steps undone in reverse: undo **op**, undo reading the front, then **snoc**; or undo **embed**, then **wrap**

(e) undoing the read of the front while keeping the last digit is **val**<sup>°</sup> on the front alone

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| $\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$<br>$\text{op}^\circ$ splits a number into its quotient and remainder by 10<br>$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$<br>$\text{op}^\circ$ gives a pair with a nonzero first component exactly at the numbers with two or more digits<br>$(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$<br>$\text{embed}^\circ$ gives a digit exactly at the one-digit numbers | (6.1b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|

## §6.2. Least fixed points

(6.2a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$  .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$  .

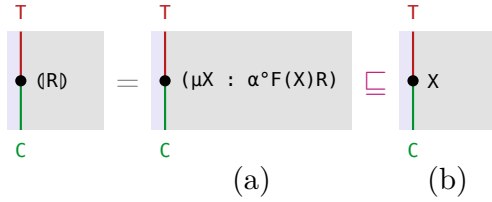
(6.2b)

| the law                                                                                    | what it says                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §Section 6.3 and every chapter after it uses                                      |
| $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |

$$\alpha^\circ F(X)R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$$

(6.2c)

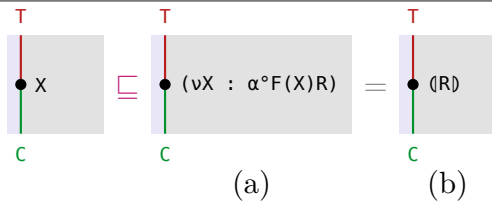
a relation  $X$  that contains one unfolding of the fold's recursion at  $X$  contains the fold



$$X \sqsubseteq \alpha^\circ F(X)R \Rightarrow X \sqsubseteq \langle R \rangle$$

(6.2d)

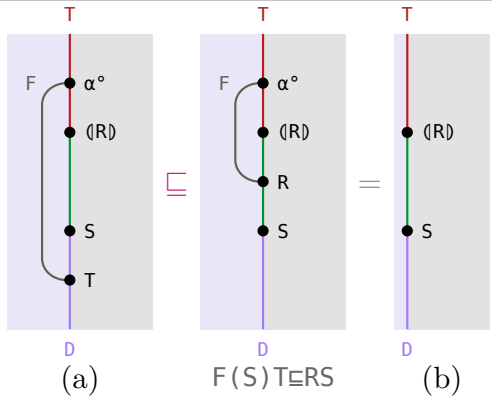
a relation  $X$  contained in one unfolding of the fold's recursion at  $X$  is contained in the fold



$$F(S)T \sqsubseteq RS \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2e)

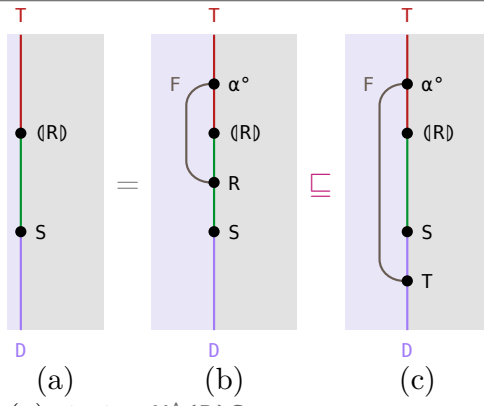
if  $S$  followed by  $R$  absorbs  $T$  after  $F(S)$ , the fold of  $T$  is below the fold of  $R$  followed by  $S$



$$RS \sqsubseteq F(S)T \Rightarrow (R)S \sqsubseteq (T)$$

(6.2f)

if  $R$  followed by  $S$  is absorbed by  $F(S)$  followed by  $T$ , the fold of  $R$  followed by  $S$  is below the fold of  $T$



(a) (6.3) at  $X \triangleq (R)S$

(b)  $\alpha(R) = F((R))R$ ,  $\alpha^0 \alpha = 1$  —  $\triangleq$  reduce

(c)  $RS \sqsubseteq F(S)T$

$F(R)F(S) = F(RS)$  —  $\triangleq$  relator

### §6.3. Hylomorphisms

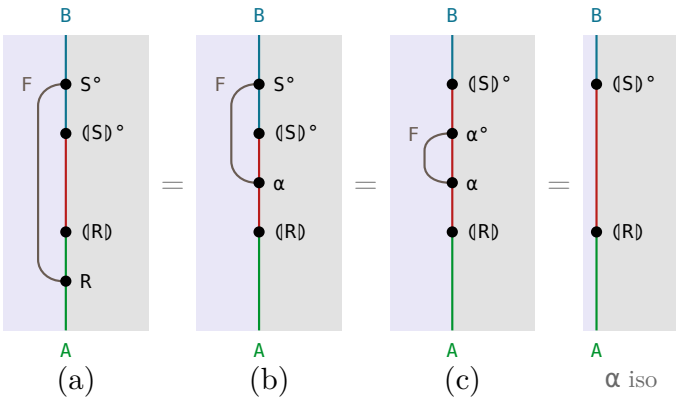
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : \mathbb{S}^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$\mathbb{S}^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term  $\mathbb{S}^\circ$  represents the decomposition stage,  $F((\mathbb{S})^\circ(\mathbb{R}))$  the stage of solving the subproblems recursively, and  $R$  the recombination stage;  $R : FA \rightarrow A$ ,  $\mathbb{S} : FB \rightarrow B$ ,  $\alpha : FT \rightarrow T$  initial



(a) the body at  $(\mathbb{S})^\circ(\mathbb{R})$

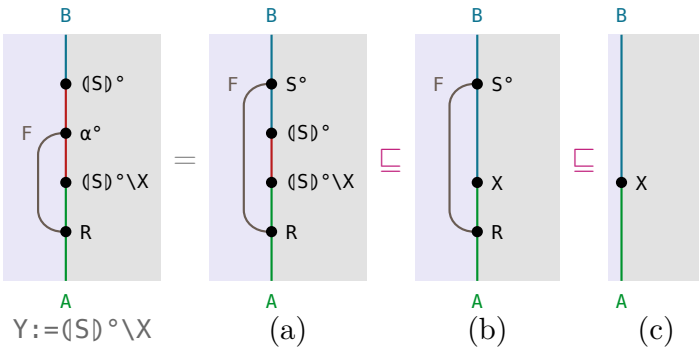
$$F(RS) = F(R)F(S) \text{ — } \triangleq \text{relator}$$

(b)  $F((\mathbb{R}))R = \alpha(\mathbb{R}) \text{ — } \triangleq \text{reduce}$

(c)  $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ) \text{ — } \triangleq \text{reduce, relator-laws}$

$$\mathbb{S}^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism  $(\mathbb{S})^\circ(\mathbb{R})$  is included in  $X$  if  $X$  satisfies the associated recursion inequation

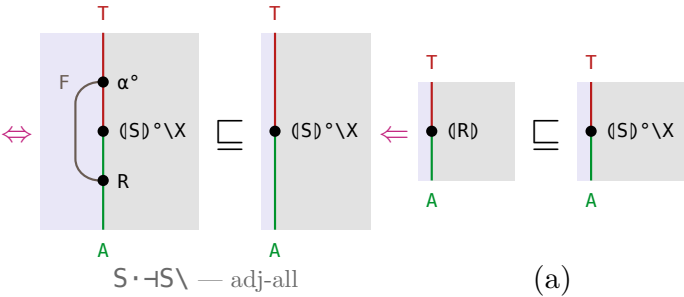


(a)  $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ)$

$$F(RS) = F(R)F(S) \text{ — } \triangleq \text{relator}$$

(b)  $(\mathbb{S})^\circ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \text{ — adj-all}$

(c)  $\mathbb{S}^\circ F(X)R \sqsubseteq X$



$\mathbb{S} \cdot \mathbb{S} \setminus \text{ — adj-all}$

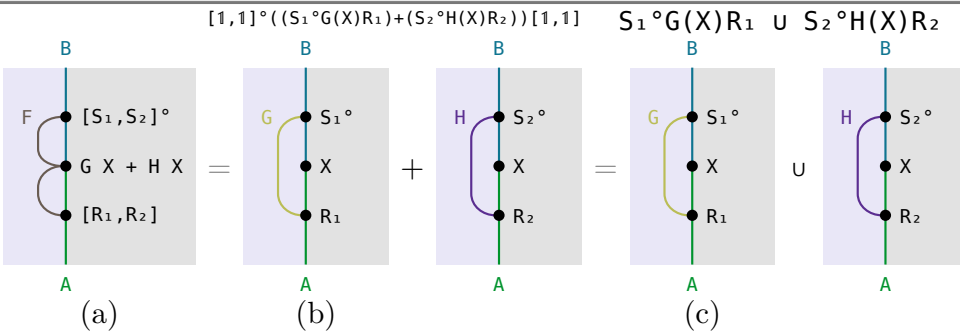
(a)

(a)  $(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \text{ — } \triangleq \text{reduce, (6.2b), adj-all}$

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \implies ([S_1, S_2])^\circ([R_1, R_2]) = (\mu X : S_1^\circ G(X)R_1 \cup S_2^\circ H(X)R_2)$$

(6.3b)

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



(a) the body of Theorem 6.2 at  $F(X) = G(X) + H(X)$

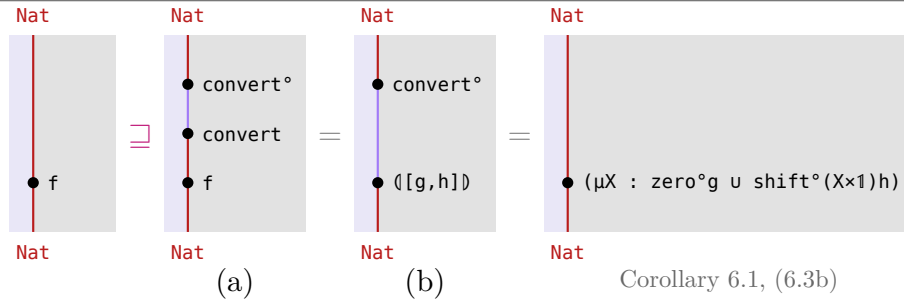
(b)  $[S, T] = (S + T) \nabla$  on both sides gives  $\nabla^\circ (S_1^\circ G(X)R_1 + S_2^\circ H(X)R_2) \nabla$ ; the  $+$  stands for the codiagonals  $\nabla^\circ, \nabla$ , which are not drawn

(c)  $\nabla^\circ (P + Q) \nabla = P \cup Q \text{ — coproduct}$

## §6.4. Fast exponentiation and modulus computation

$$\text{zero } f=g \wedge \text{shift } f=(f \times 1)h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1)h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



(a) `convert` simple

(b) fusion: `zero f=g`, `shift f=(f×1)h`

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

`zero exp(a)=one`

`a` to the power 0 is 1

`shift exp(a)=(exp(a)×1)op(a)`

`a` to the power `2n+d` is `op(a)` of `a` to the power `n` and `d`

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

`zero mod(b)=zero`

0 mod `b` is 0

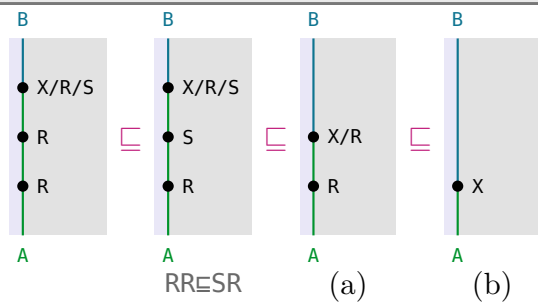
`shift mod(b)=(mod(b)×1)op(b)`

`(2a+d) mod b` is `op(b)` of `a mod b` and `d`

## §6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \sqsubseteq SR \Rightarrow \text{inductive}(R) \quad (6.5a)$$

**R** is inductive when **R**  $\sqsubseteq$  **S****R** for an inductive **S**



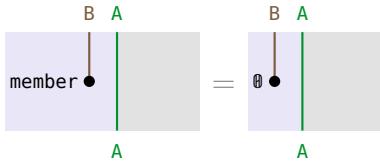
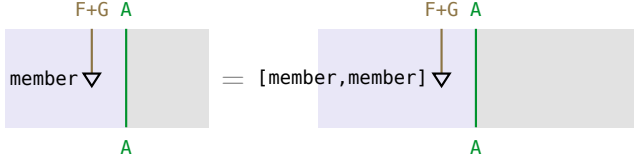
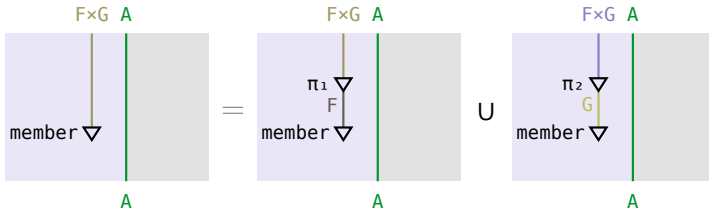
(a) division by `S`

(b) division by `R`

**member= $\emptyset$** 

(6.5b)

a constant relator records no elements, so its membership is empty

**member=[member, member]**a member of  $F+G$  is a member of whichever summand the value is in**member= $\pi_1$  member  $\cup$   $\pi_2$  member**a member of  $F \times G$  is a member of either component**inductive(Smember)  $\wedge$   $X=SF(X)R \wedge Y=SF(Y)R \Rightarrow X=Y$** 

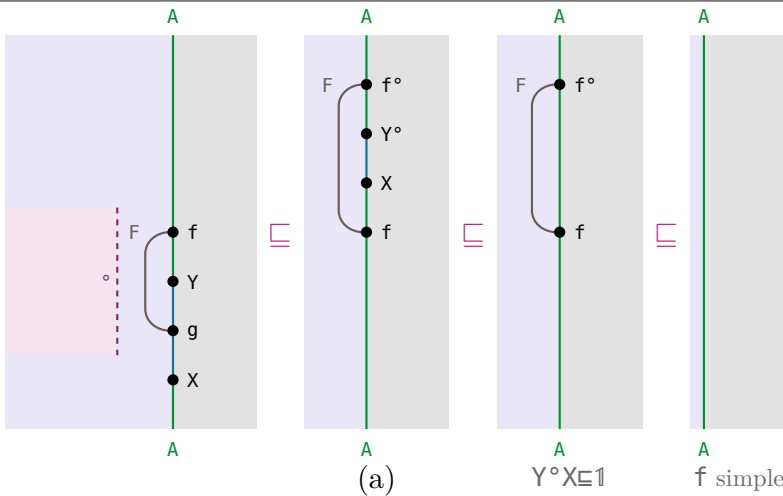
(6.5c)

when  $S$  member is inductive, two solutions  $X, Y$  of  $X=SF(X)R$  are equal**inductive(Smember)  $\wedge$  Entire( $S$ )  $\wedge$  Entire( $R$ )  $\wedge$   $SF(X)R \subseteq X \Rightarrow$  Entire( $X$ )**

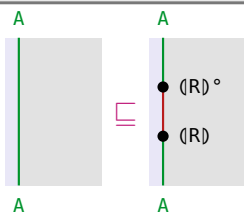
(6.5d)

when  $S$  member is inductive and  $S, R$  are entire, a pre-fixed point of  $X \mapsto SF(X)R$  is entire**inductive(gmember)  $\wedge$  Map( $g$ )  $\wedge$  Map( $f$ )  $\wedge$   $X=gF(X)f \Rightarrow$  Map( $X$ )**

(6.5e)

when  $g$  member is inductive and  $f, g$  are maps, the solution of  $X=gF(X)f$  is a map(a)  $X=gF(X)f$ ,  $g$  simple**inductive( $R^o$ member)  $\wedge$   $1 \in R^o R \Rightarrow 1 \in (R)^o (R)$** 

(6.5f)

when  $R^o$ member is inductive and  $R$  is surjective, the fold  $(R)$  is surjective

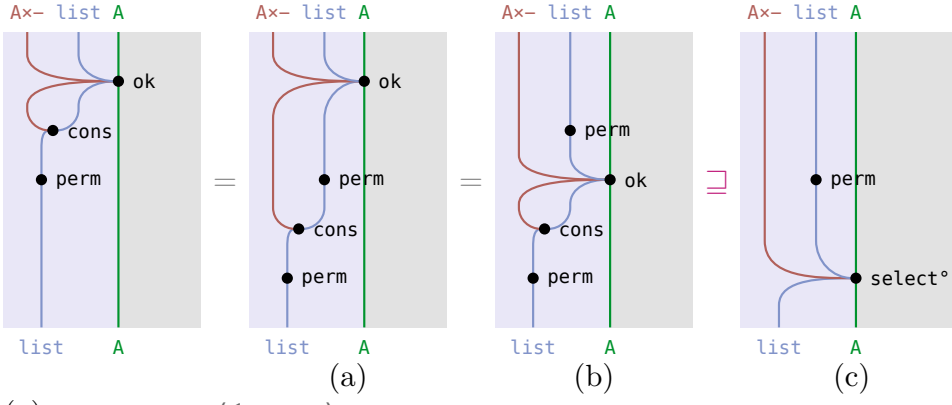
(a)

(a) (6.5d) at  $S=R^o$



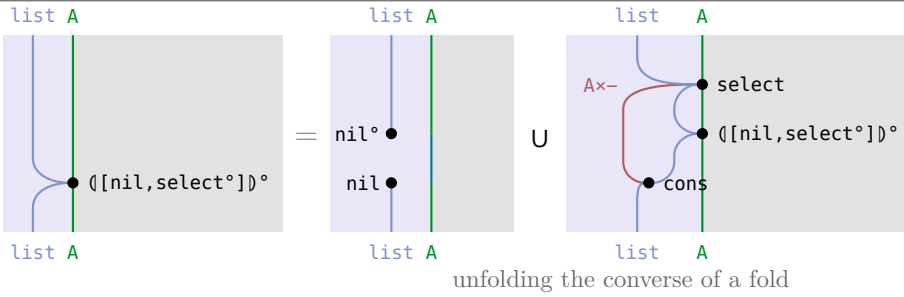
$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \cup \text{select} (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

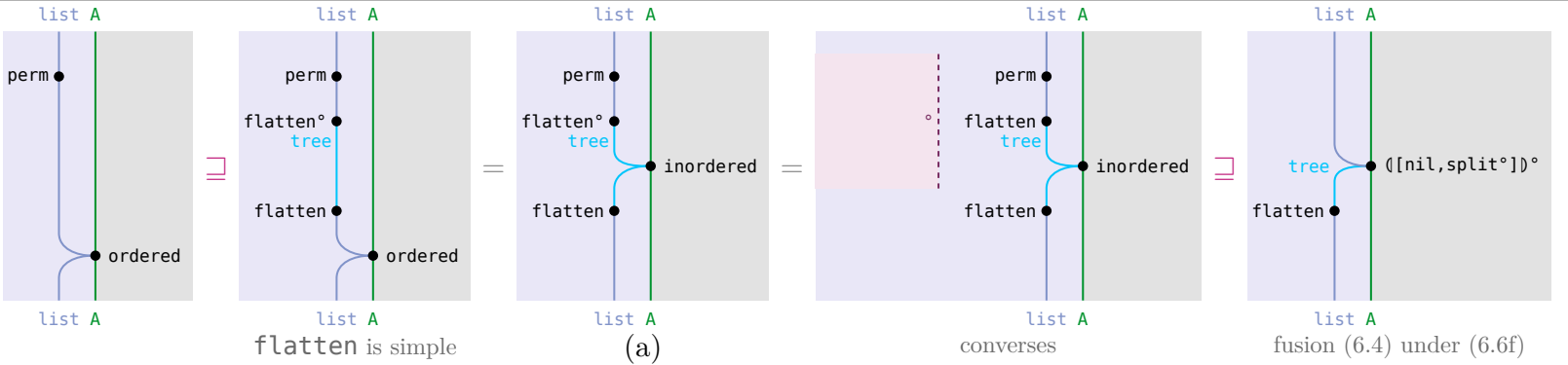
selection sort returns [] on [], and otherwise selects (a,y), sorts y and conses a back on



$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow \quad (6.6e)$$

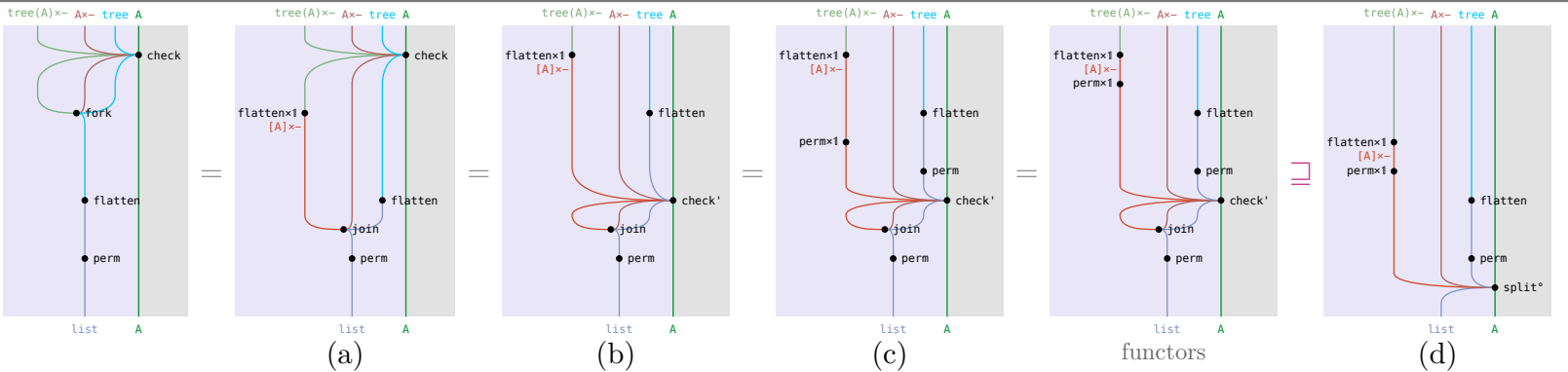
$$([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered}$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check}' \text{ fork flatten perm} \quad (6.6f)$$

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



(6.6g)

$$\llbracket \text{nil}, \text{split}^\circ \rrbracket^\circ \circ \text{flatten} = \text{nil}^\circ \text{nil} \cup \text{split}$$

$$(((\llbracket \text{nil}, \text{split}^\circ \rrbracket^\circ \circ \text{flatten}) \times (1 \times (\llbracket \text{nil}, \text{split}^\circ \rrbracket^\circ \circ \text{flatten})))) \text{join}$$

quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a

(a)  $F(X)[\text{nil}, \text{join}] = [\text{nil}, (X \times 1 \times X) \text{join}]$

## §6.7. Closure

(6.7a)

$$R^* = (\mu X : 1 \cup XR)$$

closing  $R$  by composing it on the left and closing it by composing on the right give the same relation  $R^*$

(6.8), Ex 6.31

(6.7b)

$$1 \sqsubseteq X \wedge XX \sqsubseteq X \implies R \sqsubseteq X \iff R^* \sqsubseteq X$$

for a preorder  $X$ ,  $X$  contains  $R$  exactly when it contains  $R^*$

(6.7c)

$$1 \sqsubseteq R^*$$

$R^*$  is reflexive

(a) fixed point, Thm 6.1

(6.7d)

$$R \sqsubseteq R^*$$

$R^*$  contains  $R$

identity  
(a)  $1 \sqsubseteq R^*$  — (6.7c)  
(b) fixed point

(6.7e)

**$R^*R^* \sqsubseteq R^*$**

$R^*$  is transitive, because  $R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup RX$

(a) composition distributes over  $\cup$

(b)  $RR^* \sqsubseteq R^*$  — (6.7d)

(6.7f)

**$1 \sqsubseteq X \wedge XX \sqsubseteq X \wedge R \sqsubseteq X \implies R^* \sqsubseteq X$**

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $X \mapsto 1 \cup RX$ , so it contains  $R^*$

$1 \sqsubseteq X, R \sqsubseteq X$        $XX \sqsubseteq X$

(6.7g)

**$\text{inductive}(R) \wedge X = 1 \cup RX \implies X = R^*$**

when  $R$  admits induction,  $R^*$  is the only  $X$  equal to  $1 \cup RX$

(a)  $(R^* = R^*/X) \cap R^* \sqsubseteq R^*/X$

$X = 1 \cup RX$       composition distributes over  $\cup$        $(Z/R)R \sqsubseteq Z, R^*R \sqsubseteq R^*$

(6.7h)

**$\frac{R^*}{3} = (\frac{1}{3}, \frac{RR^*}{3}) \text{ cup}$**

the set of  $R^*$ -successors of  $a$  is  $a$  itself joined with the  $R^*$ -successors of its  $R$ -successors

(a)  $\Lambda(R \cup S) = (\Lambda(R), \Lambda(S)) \text{ cup}$

$R^* = 1 \cup RR^*$

(6.7i)

**$R - \emptyset = R$**

taking nothing away leaves  $R$

**$R \cup (S - R) = R \cup S$**

beside  $R$ , only the part of  $S$  outside  $R$  adds anything

**$R - (S \cup T) = R - S - T$**

taking away  $S$  and then  $T$  takes away  $S \cup T$

**$(R \cup S) - T = (R - T) \cup (S - T)$**

taking  $T$  away from a union takes it away from each part

(6.7j)

**$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$**

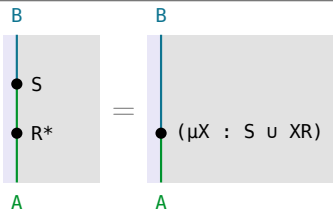
the least fixed point of  $\varphi$  after  $\psi$  is  $\varphi$  applied to the least fixed point of  $\psi$  after  $\varphi$

Ex 6.35

$$SR^* = (\mu X : S \cup XR)$$

(6.7k)

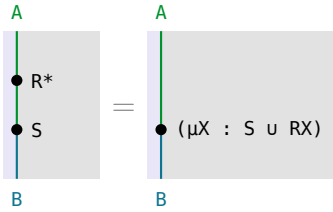
$S$  followed by any number of  $R$  steps is the least  $X$  containing  $S$  and closed under a further  $R$  step



p.160

$$R^*S = (\mu X : S \cup RX)$$

any number of  $R$  steps followed by  $S$  is the least  $X$  containing  $S$  and closed under an  $R$  step in front

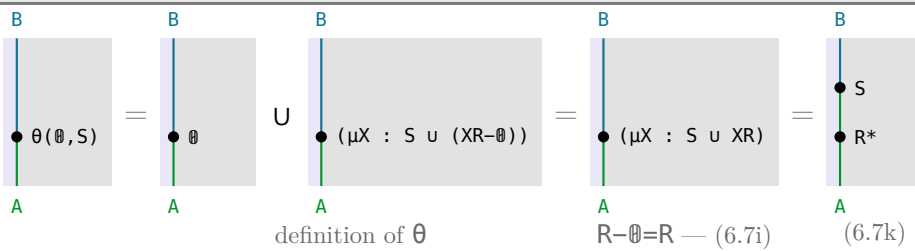


Ex 6.32

$$\theta(\emptyset, S) = SR^*$$

(6.7l)

started with nothing found and  $S$  to explore,  $\theta$  computes  $SR^*$

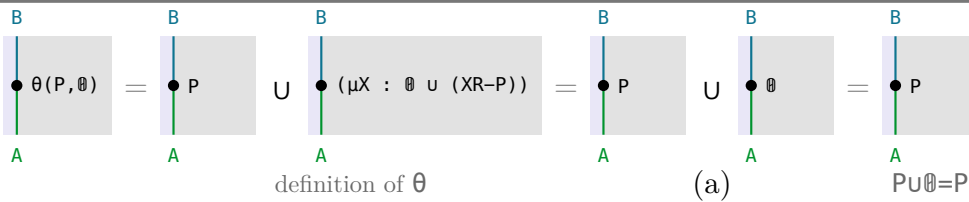
definition of  $\theta$  $R - \emptyset = R$  — (6.7i)

(6.7k)

$$\theta(P, \emptyset) = P$$

(6.7m)

with nothing left to explore,  $\theta$  returns what it has found

definition of  $\theta$ 

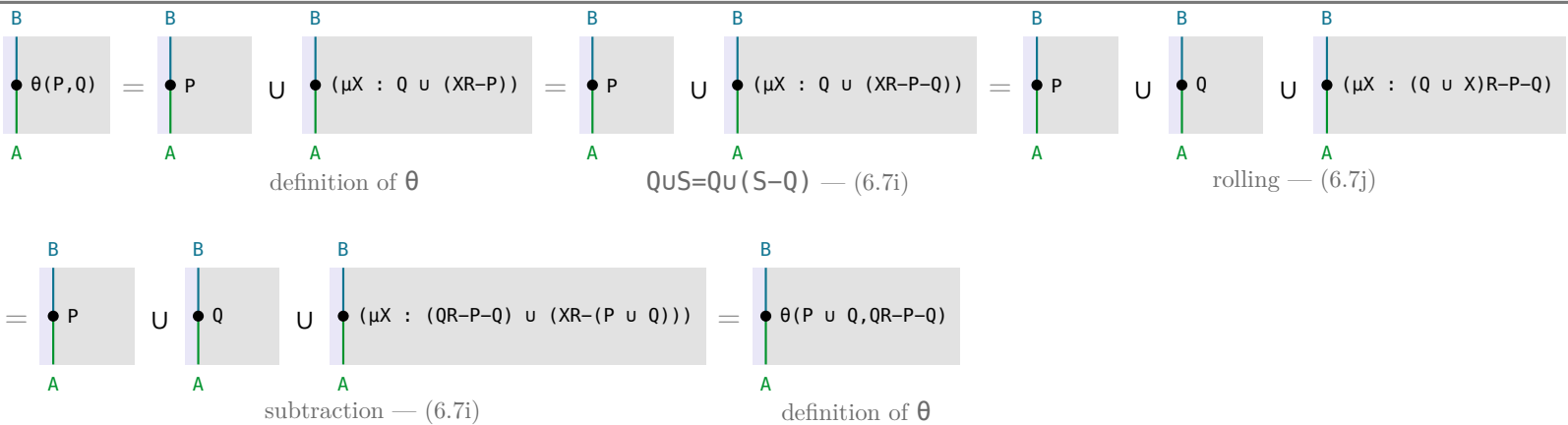
(a)

 $P \cup \emptyset = P$ (a)  $\emptyset$  is a prefixed point:  $\emptyset R - P = \emptyset$ 

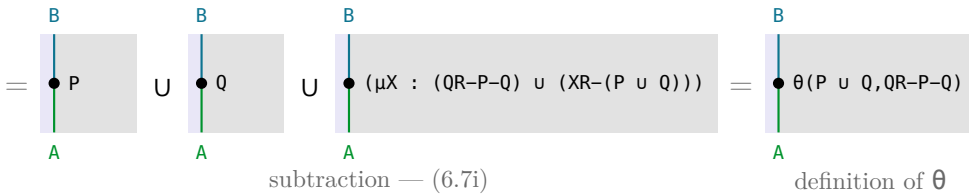
$$\theta(P, Q) = \theta(P \cup Q, QR - P - Q)$$

(6.7n)

one round moves  $Q$  into the found part and explores the new  $R$ -successors of  $Q$  that are in neither

definition of  $\theta$  $QU S = QU(S - Q)$  — (6.7i)

rolling — (6.7j)



subtraction — (6.7i)

definition of  $\theta$